

Robust Saliency-Based Speed Sensorless Control of Induction Machines under Overload Operation

E. Rodriguez Montero, M. Vogelsberger, T. Wolbank

Abstract – The stable vector control at very low speeds and standstill is only possible using saliency extraction methods. There exist specific designs that lead to induction machines having a significant rotor slotting saliency. The slotting signal, in interaction with the magnetic saturation saliency signal, generates unavoidably a further harmonic, often referred as intermodulation harmonic. In this sense, some motors can show three main harmonics: saturation, slotting and intermodulation. In available literature, intermodulation has been treated as a distortion and thus compensated using filtering or feed-forward methods. Yet, this paper proposes to employ the intermodulation saliency harmonic itself as control signal. Applying a voltage step excitation strategy to an induction motor, an intermodulation saliency vector will be acquired using the resulting phase current derivatives and signal processing. Experimental results measured on an induction motor will show that the intermodulation saliency harmonic can deliver a robust and very accurate rotor position, especially when operated in the overload range.

Index Terms– Induction motors, Intermodulation distortion, Motor drives, Sensorless control.

I. NOMENCLATURE

i	Current
U, E	Voltage, Back-EMF
L	Transient leakage inductance
T	Torque
ψ_R	Rotor flux
$\vec{S}_v$	Saliency vector
$\vec{I}_v$	Intermodulation saliency vector
$\vec{SLOT}_v$	Slotting saliency vector
$\vec{SAT}_v$	Saturation saliency vector
R_s	Stator resistance
ω_e, ω_R	Electrical speed, rotor speed
$\theta_{ele}, \theta_{mech}$	Electrical speed, mechanical speed
N_{slot}, N_{sal}	Number of rotor slots, Number of saliencies
IM	Induction machine
L_0	Transient leakage inductance offset
$L_m, \omega_m, \theta_{0,m}$	Saliency-modulated transient leakage inductance (amplitude, speed, phase shift)
a	Complex operator
LUT	Look-Up Table
FOC	Field-oriented control
<i>Subscripts</i>	
A, B, C	Three phase system
DC	Direct current
$000, 110, 010, 011, 001, 101, 111, 000$	Switching states (2-level 3-phase inverter)

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II. INTRODUCTION

THE stable control of AC drives at standstill and very low frequencies can only be guaranteed by methods that extract machine saliencies. This is because at very low frequencies, the voltage applied by the control scheme through the inverter to the machine terminals is very low, and thus fundamental wave models become unstable due to the effect of inverter non-linear switching [1].

Machine saliencies appear under specific rotor and stator geometric and magnetic designs, and therefore they contain information about the rotor position or flux. Two common saliencies in induction machines are the rotor slotting saliency and the magnetic saturation saliency. The saturation saliency is produced by the unevenness of the saturation level of the machine lamination, and its frequency is twice the electrical one. The slotting of the stator and rotor lamination causes the slotting saliency, and its frequency is N_{slot} times the mechanical one. The interaction of the slotting and saturation saliencies generates a further saliency often denoted in literature as intermodulation saliency [2,10], which appears in the system due to the non-linear behavior of the resulting transient leakage inductance that results after an inverter-imposed voltage step, when multiple saliencies are present.

A saliency vector $\vec{S}_v$ can be described as a vector where all machine saliencies are superposed. Its acquisition can be performed using combinations of phase current derivatives that result from specific voltage steps [2,6,10]. Once acquired, the $\vec{S}_v$ must be processed in order to remove all unwanted non-control saliencies. This can be done either in a feedforward scheme [3,6,9,10], after a commissioning to identify non-control saliency characteristics, or by filtering methods [4,5,7,8]. The result from both strategies is a single saliency vector (control saliency) from which rotor information can be extracted. In the available literature, signal processing for multiple-saliency machines has never aimed to extract the intermodulation saliency and use it as the sensorless control signal. Rather, saturation [6-9] or slotting [2-5,10] have been the goal of the saliency extraction. Whenever intermodulation was present in literature, it was either treated as a distortion [1-3,10] or not mentioned. This paper proposes the novel concept of employing the intermodulation saliency harmonic

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as the control signal. Applying a voltage step excitation strategy to an induction machine, consisting of the three sets of antiparallel switching states, an intermodulation saliency vector $\bar{I}_v$ will be acquired using the resulting current derivatives and specific signal processing. Experimental results will prove that the intermodulation saliency harmonic is highly suitable for sensorless control as it delivers a robust and very accurate rotor position for all load levels.

III. SALIENCY EXTRACTION

Fig.1 shows an equivalent transient circuit of the IM, fed by a two-level three-phase inverter. Within the inverter, the eight SVPWM inverter states in digital form are shown.

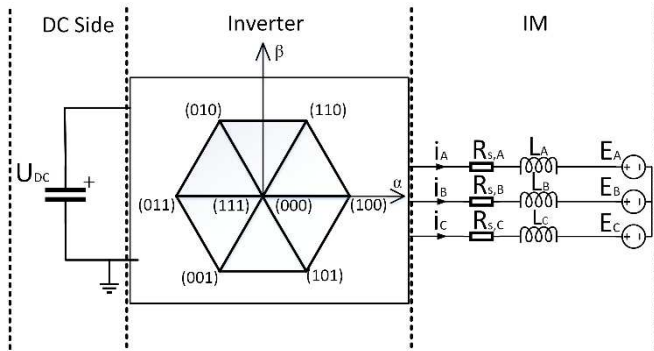


Fig. 1. DClink-inverter-motor diagram.

The modulation of the air-gap flux due to the machine saliencies is reflected at L_A , L_B and L_C , which can be accessed using voltage step excitations and calculating the current slopes on the measured phase current responses. Assume 100 and 011 is switched by the inverter and their resulting current slopes are calculated. According to Fig.1, (1)-(6) hold true:

$$U_{DC} = R_{s,A} \cdot i_{A,100} + L_A \cdot di_{A,100}/dt + E_{A,100} - R_{s,B} \cdot i_{B,100} - L_B \cdot di_{B,100}/dt - E_{B,100} \quad (1)$$

$$U_{DC} = R_{s,A} \cdot i_{A,100} + L_A \cdot di_{A,100}/dt + E_{A,100} - R_{s,C} \cdot i_{C,100} - L_C \cdot di_{C,100}/dt - E_{C,100} \quad (2)$$

$$0 = R_{s,B} \cdot i_{B,100} + L_B \cdot di_{B,100}/dt + E_{B,100} - R_{s,C} \cdot i_{C,100} - L_C \cdot di_{C,100}/dt - E_{C,100} \quad (3)$$

$$-U_{DC} = R_{s,A} \cdot i_{A,011} + L_A \cdot di_{A,011}/dt + E_{A,011} - R_{s,B} \cdot i_{B,011} - L_B \cdot di_{B,011}/dt - E_{B,011} \quad (4)$$

$$-U_{DC} = R_{s,C} \cdot i_{C,011} + L_C \cdot di_{C,011}/dt + E_{C,011} - R_{s,A} \cdot i_{A,011} - L_A \cdot di_{A,011}/dt - E_{A,011} \quad (5)$$

$$0 = R_{s,B} \cdot i_{B,011} + L_B \cdot di_{B,011}/dt + E_{B,011} - R_{s,C} \cdot i_{C,011} - L_C \cdot di_{C,011}/dt - E_{C,011} \quad (6)$$

Where the transient leakage inductance can be expressed as (7-10):

$$L_A = L_0 + L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m}) \quad (7)$$

$$L_B = L_0 + L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m} + 2 \cdot \pi/3) \quad (8)$$

$$L_C = L_0 + L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m} + 4 \cdot \pi/3) \quad (9)$$

Note that for clarity reasons, only a single saliency is represented by (7)-(9), although there usually exist multiple saliencies. Rearranging and solving (1)-(6), with the help of (7)-(9) a current slope difference can be created:

$$\frac{di_{A,100}}{dt} - \frac{di_{A,011}}{dt} = \frac{2 \cdot L_0 - L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m})}{cst} \quad (10)$$

$$\frac{di_{B,100}}{dt} - \frac{di_{B,011}}{dt} = \frac{-L_0 - L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m} + 4 \cdot \frac{\pi}{3})}{cst} \quad (11)$$

$$\frac{di_{C,100}}{dt} - \frac{di_{C,011}}{dt} = \frac{-L_0 - L_m \cdot \cos(\omega_m \cdot t + \theta_{0,m} + 2 \cdot \frac{\pi}{3})}{cst} \quad (12)$$

$$cst = (3 \cdot L_0^2 - 3/4 \cdot L_m^2)/(2 \cdot U_{DC}) \quad (13)$$

$$\begin{aligned} \frac{di_{ABC,100}}{dt} - \frac{di_{ABC,011}}{dt} &= \frac{di_{A,100}}{dt} - \frac{di_{A,011}}{dt} \\ &+ \left(\frac{di_{B,100}}{dt} - \frac{di_{B,011}}{dt} \right) \cdot a \\ &+ \left(\frac{di_{C,100}}{dt} - \frac{di_{C,011}}{dt} \right) \cdot a^2 \end{aligned} \quad (14)$$

Doing the same calculations for the 010-101 and 001-110 inverter states, the following saliency vector can be obtained:

$$\begin{aligned} \bar{S}_v &= \left(\frac{di_{ABC,100}}{dt} - \frac{di_{ABC,011}}{dt} \right) \\ &+ \left(\frac{di_{ABC,010}}{dt} - \frac{di_{ABC,101}}{dt} \right) \cdot a \\ &+ \left(\frac{di_{ABC,001}}{dt} - \frac{di_{ABC,110}}{dt} \right) \cdot a^2 \\ &= \frac{-\frac{9}{2} \cdot L_m \cdot e^{j \cdot (\omega_m \cdot t + \theta_{0,m})}}{cst} \end{aligned} \quad (15)$$

IV. INTERMODULATION SALIENCY AND ITS SEPARATION

If the input to a non-linear system is the sum of two or more sinusoids (see (1-12)), a third component named intermodulation is unavoidably generated.

As stated previously, an induction machine can have multiple saliencies, all of them appearing superposed in the vector $\bar{S}_v$. For some induction machines, dominant saliencies are slotting and saturation. Because of both saliencies and the non-linear system shown in (1-6), an intermodulation saliency harmonic is generated. This additional harmonic has a different amplitude and speed than the two harmonics whose interaction created the intermodulation. Yet, it shows a deterministic behavior. In general, each saliency speed can be described as a combination of electrical and mechanical speed ω_E and ω_R , as in (16).

$$\omega_m = N_E \cdot \omega_E + N_R \cdot \omega_R \quad (16)$$

The frequencies of each saliency are listed in Table I:

	N_E	N_R
$\bar{SLOT}_v$	0	$-N_{slot}$
$\bar{SAT}_v$	2	0
$\bar{I}_v$	-2	N_{slot}

Fig. 2 shows an example of $\bar{S}_v$ where the three saliencies of Table I are present. It is taken from the motor presented in the experimental results section.

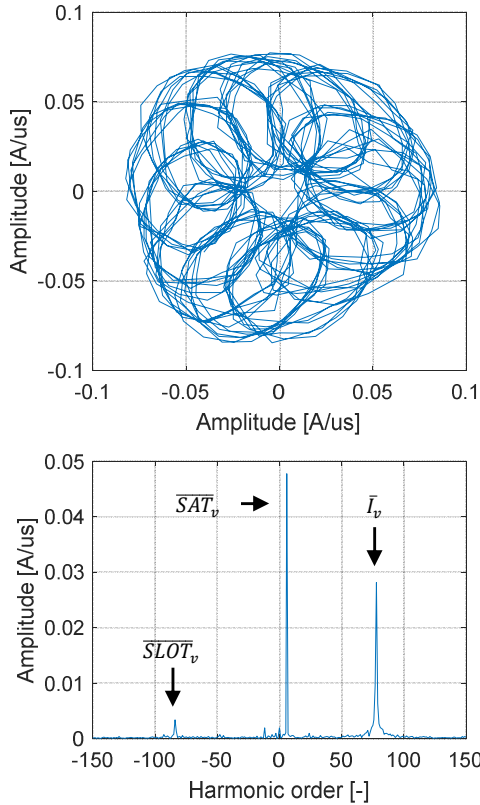


Fig. 2. Example of calculated $\bar{S}_v$ during one rotor revolution at no-load (top) and its symmetrical FFT (bottom).

As Fig. 2 shows, three saliency harmonics form the $\bar{S}_v$. From left to right, these are slotting, saturation and intermodulation. The amplitude of the intermodulation saliency increases as the torque increases. This is because the saturation harmonic increases as the torque increases. This can be seen in Fig. 3, where the saliency amplitudes of $\bar{S}_v$ are plotted for three different torque levels. The results are obtained from the induction machine described in section VI.

As it is clearly observed in Fig. 3, the amplitude of $\bar{SLOT}_v$ is significantly smaller than those from $\bar{SAT}_v$ and $\bar{I}_v$, which implies that the extraction of $\bar{SLOT}_v$ is getting extremely difficult when loading the machine.

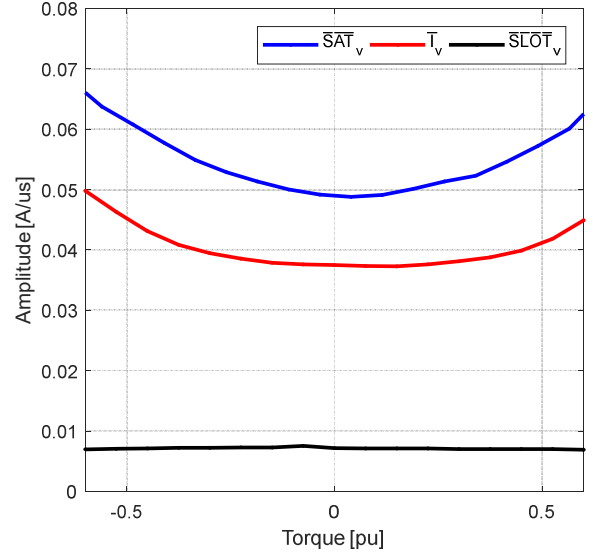


Fig. 3. Calculated amplitude of $\bar{SLOT}_v$, $\bar{SAT}_v$, $\bar{I}_v$ for different torque levels.

In this sense, the intermodulation saliency can be alternatively extracted, as within this saliency, rotor position information is contained. This paper proposes to extract $\bar{I}_v$ by means of feedforward compensating the saturation and slotting saliencies. Besides, signal processing is used to compute a rotor position from the angle extracted from $\bar{I}_v$.

During a commissioning, the saliency vector amplitudes and phase shifts are calculated and stored in a *LUT* [10]. The amplitude was shown in Fig. 3. Stored information is then accessed during online operation in order to eliminate all non-control signals. In this paper, saturation and slotting are compensated from the $\bar{S}_v$ in a feedforward scheme.

V. BLOCK DIAGRAM OF PROPOSED DRIVE

The proposed drive is based on the *FOC* of an induction machine where the rotor flux is calculated by means of the stator current phasor and the rotor position, using the rotor equation in rotor-fixed coordinates. As explained before, voltage step excitation is applied regularly to the machine and the current response measured. The current slopes, calculated using a least-square linear regression, are processed as explained in Section III. The result of the current slope combinations, as explained by (10)-(15), is the saliency vector $\bar{S}_v$. In $\bar{S}_v$, all saliencies are superposed. Using feedforward compensation, non-control saliencies (saturation, slotting) are

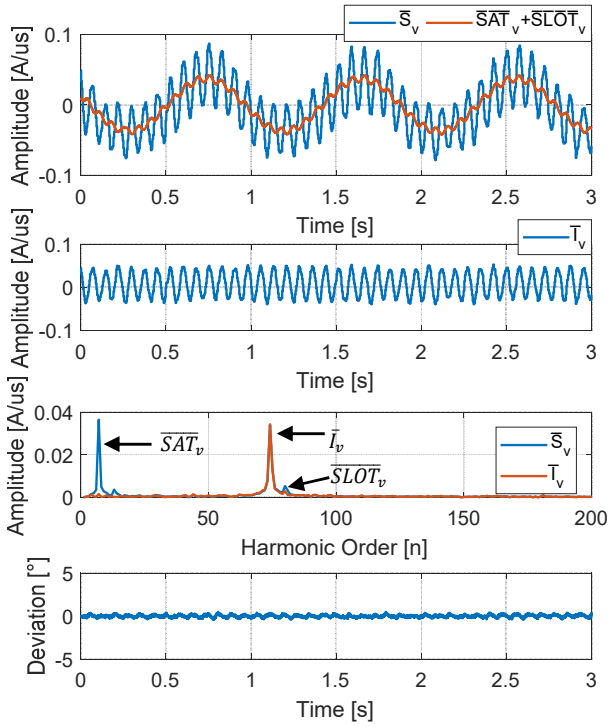


Fig. 5. 1st plot: real component of calculated $\bar{S}_v$ (blue) and sum of identified $\bar{SLOTV}_v$ and $\bar{SAT}_v$ during commissioning (red).
2nd plot: real component of $\bar{I}_v$ resulting from feedforward elimination of $\bar{SLOTV}_v$ and $\bar{SAT}_v$ from $\bar{S}_v$.
3rd plot: FFT of real components of $\bar{S}_v$ and $\bar{I}_v$.
4th plot: Deviation angle, calculated as encoder minus sensorless. The sensorless rotor angle is obtained from the $\bar{I}_v$ using signal processing shown in section V.

As Fig 5. shows, the $\bar{I}_v$, present in the $\bar{S}_v$, is successfully extracted for the given operating point. The extraction is done feedforward compensating the $\bar{SLOTV}_v$ and $\bar{SAT}_v$ from the $\bar{S}_v$. The result of this saliency feedforward compensation is the $\bar{I}_v$, which has been always treated as a distortion in available literature [1-3,10]. However, this paper uses the $\bar{I}_v$ as the sensorless control signal, as it inherently contains the mechanical and electrical frequencies within it. Using the proposed novel signal processing of Fig. 4, the rotor position is acquired with very high precision.

C. Variable load and speed test

During the following experiment, the IM under test was controlled at variable torque and rated magnetization current. The speed was controlled by the coupled machine at different set-points. The experimental test at variable load and speed is shown in Fig. 6.

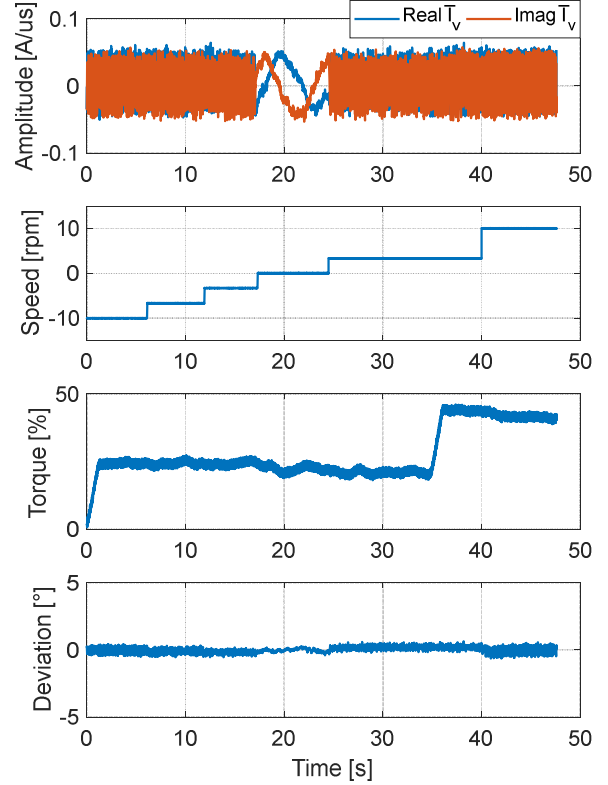


Fig. 6. 1st plot: real and imaginary components of $\bar{I}_v$.
2nd plot: ω_R .
3rd plot: Measured T.
4th plot: Deviation angle, calculated as encoder minus sensorless. The sensorless rotor angle is obtained from the $\bar{I}_v$ using signal processing shown in section V.

As Fig. 6 shows, the novel concept of employing $\bar{I}_v$ as the sensorless control signal achieved an outstanding rotor angle precision in the considered speed and torque range. By feedforward compensating the $\bar{SLOTV}_v$ and $\bar{SAT}_v$ from the $\bar{S}_v$, a robust $\bar{I}_v$ was obtained during the whole experiment. The angle of the $\bar{I}_v$ was transformed into a sensorless rotor position using the novel signal processing of Fig. 4. The deviation angle, calculated as the encoder position minus the sensorless rotor angle, remains within $\pm 0.7^\circ$ during the whole measurement, thus proving excellent control performance.

VII. CONCLUSIONS

Depending on the machine design, induction machines develop rotor related saliency in addition to the saturation related saliency. The interaction of these two saliencies generates another saliency, often named as intermodulation. It contains information about both rotor slotting and magnetic saturation and increases its amplitude with increasing load. In the available literature, intermodulation saliency has been compensated by feedforward methods or filtered, since it was considered a distortion. For the first time, this work employed the intermodulation saliency as the control signal. By means of a voltage step excitation scheme and specific current slope processing, a saliency vector was computed. Using feedforward compensation, the intermodulation saliency was successfully separated from other non-control saliencies. The

provided novel signal processing permitted the robust acquisition of the intermodulation saliency and the transformation of the intermodulation saliency angle into an outstanding rotor position signal. A test bench with an induction machine having slotting, saturation and intermodulation saliencies was investigated. The experimental results show that the novel intermodulation saliency extraction method and processing is highly suitable for sensorless control, resulting in very robust and accurate rotor position information, especially when considering loaded operation.

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X. BIOGRAPHIES

E. Rodriguez Montero (Student Member, IEEE) was born in León, Spain, in 1993. He received the B.Sc. in electromechanical engineering from the Pontificia Comillas University, Madrid, Spain, in 2015, and the M.Sc. in electrical engineering from the Denmark Technical University, Copenhagen, Denmark, in 2017.

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